

Lab 03 Array of objects and More on classes

- Q1 Array of Day objects
- Q2 Array of Day objects - more operations
- Q3 Allocation of Tasks to Teams (need redo without reading hints)
- Q4 [Homework] Extension of Q3

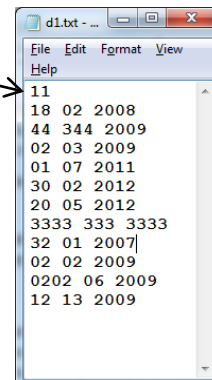
Q1. You are given a program to finish (Main.java, Day.java), and data files (d1.txt, d2.txt etc). The program should input a list of day entries from a file and display them, as shown below.

Note that some date values are wrong (eg. 44/344/2009). We will handle this problem in Q2.

File contents:

First line: number of day entries in the following

Each remaining line is supposed to contain the day, month, and year of a date



Sample input/output:

Please enter the filename: d1.txt

Totally 11 entries:

1. 18/2/2008
2. 44/344/2009
3. 2/3/2009
4. 1/7/2011
5. 30/2/2012
6. 20/5/2012
7. 3333/333/3333
8. 32/1/2007
9. 2/2/2009
10. 202/6/2009
11. 12/13/2009

Underlined content is input by the user

Please complete the program (see Given_Code.txt), test it.
Submit both Main.java and Day.java to PASS.

Draw the array and Day objects below:

Q2. Extend the program from Q1 so that it displays the valid and invalid ones, as shown on the right:

Your tasks:

- (1) Download and finish `Main.java` for Q2.
The program should read the file only once.
(Need help? Ask!)
- (2) Copy your `Day.java` from Q1 and **add Version 2** of the `valid` method:

Please enter the filename: `c:\dl.txt`

Valid dates:

18 Feb 2008
2 Mar 2009
1 Jul 2011
20 May 2012
2 Feb 2009

Invalid dates:

44/344/2009
30/2/2012
3333/333/3333
32/1/2007
202/6/2009
12/13/2009

Version 1 (already given in `Day.java`) - a **static** method (ie. doesn't refer to an existing `Day` object)

Keep both versions in `Day.java`

```
// check if y,m,d valid
static public boolean valid(int y, int m, int d)
{
    if (m<1 || m>12 || d<1) return false;
    switch(m){
        ..//check against 31,30,28,29
    }
    return false;
}
```

Version 2 (YOUR TASK: please add) - a **non-static** method (ie. checks the calling `Day` object)

```
public boolean valid()
{
    .. //Call version 1
}
```

- (3) Test the program and then submit `Main.java` and `Day.java` to PASS.

Lab03- Progress Check

Fill in the following:

- _____: tells **the count of elements in an array**
- _____: is the type for **an array of `Day` objects**
- _____: is to **create an array of n slots**, each slot can hold **a reference to a `Day` object**
- _____: is to **create a new `Day` object** and **make `arr[i]` link to the `Day` object**.
- _____, like `Day.createDayListFromFile(..)`, **do not require an existing object to run on**.
- _____, means using **the same name for different methods**, but **parameters must be different**, so that the compiler can decide which method you will be calling. **(see: the 2 valid methods)**
- `Day.java` in Q2 provides **only** `static public boolean valid(.....)` / **only** `public boolean valid()` / **both** ^{*circle the correct one}

Choose answers from below:

`.length`
`Day[]`
`new Day[n]`
`arr[i] = new Day(.....);`
static methods
method overloading

Q3. Complete a program that performs the following tasks:

- Input a list of student names from a file (Like Q2, the first line is the count of entries).
- Allocate them to 5 teams: First 6 students to team A, next 6 students to team B, etc..
- Ask the user to input the lab task for each team: **Lab05- Lab09**
- Show the team assignments by tasks (Note - Refer to given code: no sorting is needed!!)

Do not sort!

For simplicity, here we assume that there are 30 students in the list.

Sample input/output: (Underlined content is input by the user)

Please input the file pathname: c:\Students.txt

Grouping result:

Team A: [YIM Ho] [LAU Tak Wah] [LAW Kar Ying] [CHEUNG Hok Yau] [CHAN Ho Sun] [NG Kwun Yu]
 Team B: [YIP Tung] [TO Kei Fung] [TSANG Chi Wai] [WONG Kar Wai] [TSUI Hark] [CHENG Yu Lin]
 Team C: [LEUNG Ka Fai] [YUEN Wing Yee] [LIU Kai Chi] [HUNG Kam Bo] [MUI Yim Fong] [YUEN Wah]
 Team D: [CHAN Siu Chun] [CHOW Yun Fat] [HUI On Wah] [FUNG Bo Bo] [HUI Koon Man] [WONG Chau Sang]
 Team E: [YIP Tak Han] [CHOW Sing Chi] [LAW Lan] [SIAO Fong Fong] [CHEUNG Kwok Wing] [CHEUNG Pak Chi]

Enter tasks for the teams (Lab05, Lab06, Lab07, Lab08, Lab09):

Team A: Lab07

Team B: Lab09

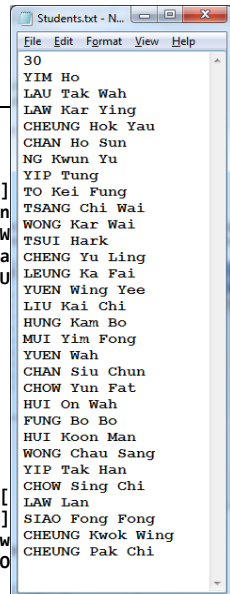
Team C: Lab05

Team D: Lab08

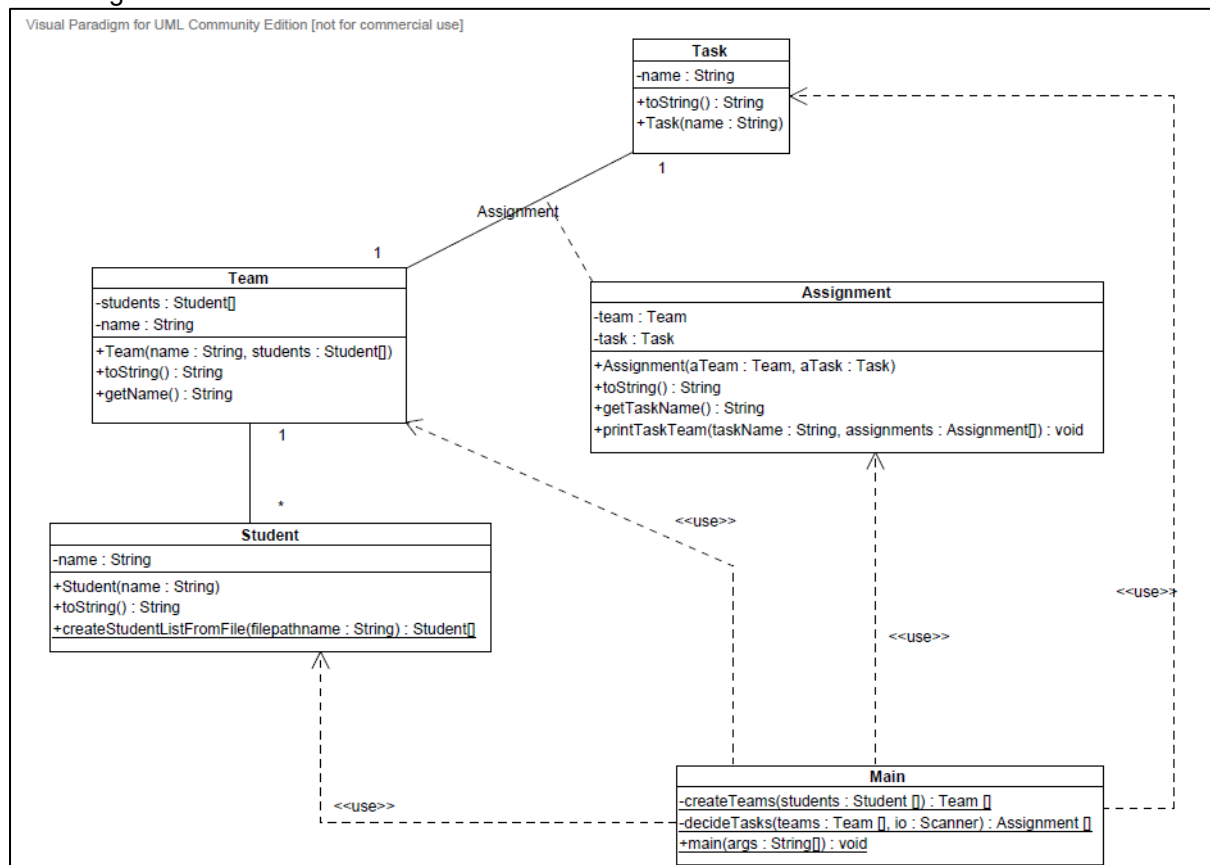
Team E: Lab06

Sorted listing by tasks:

Lab05 Team C: [LEUNG Ka Fai] [YUEN Wing Yee] [LIU Kai Chi] [HUNG Kam Bo] [MUI Yim Fong] [YUEN Wah]
 Lab06 Team E: [YIP Tak Han] [CHOW Sing Chi] [LAW Lan] [SIAO Fong Fong] [CHEUNG Kwok Wing] [CHEUNG Pak Chi]
 Lab07 Team A: [YIM Ho] [LAU Tak Wah] [LAW Kar Ying] [CHEUNG Hok Yau] [CHAN Ho Sun] [NG Kwun Yu]
 Lab08 Team D: [CHAN Siu Chun] [CHOW Yun Fat] [HUI On Wah] [FUNG Bo Bo] [HUI Koon Man] [WONG Chau Sang]
 Lab09 Team B: [YIP Tung] [TO Kei Fung] [TSANG Chi Wai] [WONG Kar Wai] [TSUI Hark] [CHENG Yu Lin]



The design:



The 5 classes are listed below:

- Task** - only the task name is stored (← Your task: finish this one **first**)
- Team** - the team name and an array of student references (← Finish this one and then the remaining classes)

- **Student** - the student name is stored; in addition there is a static method which reads a list of students from the file and returns an array of student objects *Note: the use of `.nextLine()` to take away the line break.*
- **Assignment** - a matching of team and lab task; in addition there is a static method for printing the team who is assigned with a task of the given task name.
- **Main** - the `main()` routine; in addition there are private static methods for creating the teams from the student array, and assignment of lab tasks to teams based on the user input.

Test your program and submit to PASS. Zip file can be used (but make Main.java separated):

H:\...\src\src.zip

H:\...\src\Main.java

Note: Re-do on your own, don't look at the hints/guidelines

Q4. [Take home exercise]

Must digest Q3 first

Extend your program of Q3 such that

- It allows the input file to have various number of students
- It prompts the user for the count of teams (But if it cannot divide the number of students, terminate the program. *Do the checking using %: if (students.length % numOfTeams != 0)*)
- It prompts the user for (1) the names of the tasks and (2) the task for each team.
 - * You may need an array of Strings. To get each task name, use `.next()`.
 - * To make team letters A, B, C etc.: `char teamLetter=(char)('A'+i); //i is 0, 1, 2 etc..`
 - * No checking is needed for both steps.
- **ArrayList is allowed** (If you have learnt JAVA before and you want to use JAVA's *ArrayList*, it is fine.)
- **Do NOT use StringBuilder.** *(But you may ask for special approval 😊)*

Hint

Sample rundown #1 (Underlined content is input by the user)

```
Please input the file pathname: Students16.txt
Total number of students: 16
Enter the number of teams: 3
Wrong input - It is not a factor of 16.
```

Sample rundown #2 (Underlined content is input by the user)

```
Please input the file pathname: Students24.txt ---input by the user
Total number of students: 24
Enter the number of teams: 8 ---input by the user

Grouping result:
Team A: [YIP Tung] [TO Kei Fung] [TSANG Chi Wai]
Team B: [WONG Kar Wai] [TSUI Hark] [CHENG Yu Ling]
Team C: [LEUNG Ka Fai] [YUEN Wing Yee] [LIU Kai Chi]
Team D: [HUNG Kam Bo] [MUI Yim Fong] [YUEN Wah]
Team E: [CHAN Siu Chun] [CHOW Yun Fat] [HUI On Wah]
Team F: [FUNG Bo Bo] [HUI Koon Man] [WONG Chau Sang]
Team G: [YIP Tak Han] [CHOW Sing Chi] [LAW Lan]
Team H: [SIAO Fong Fong] [CHEUNG Kwok Wing] [CHEUNG Pak Chi]

Enter 8 task names (eg. "Lab05 Lab06 Lab07 .."): Lab02 Lab04 Lab05 Lab07 Lab08 Lab09 Lab11 Lab12

Enter tasks for the teams (Lab02,Lab04,Lab05,Lab07,Lab08,Lab09,Lab11,Lab12):
Team A: Lab04
Team B: Lab07
Team C: Lab09
Team D: Lab11 ---input by the user
Team E: Lab12
Team F: Lab02
Team G: Lab08
Team H: Lab05

Sorted listing by tasks:
Lab02 Team F: [FUNG Bo Bo] [HUI Koon Man] [WONG Chau Sang]
Lab04 Team A: [YIP Tung] [TO Kei Fung] [TSANG Chi Wai]
Lab05 Team H: [SIAO Fong Fong] [CHEUNG Kwok Wing] [CHEUNG Pak Chi]
Lab07 Team B: [WONG Kar Wai] [TSUI Hark] [CHENG Yu Ling]
Lab08 Team G: [YIP Tak Han] [CHOW Sing Chi] [LAW Lan]
Lab09 Team C: [LEUNG Ka Fai] [YUEN Wing Yee] [LIU Kai Chi]
Lab11 Team D: [HUNG Kam Bo] [MUI Yim Fong] [YUEN Wah]
Lab12 Team E: [CHAN Siu Chun] [CHOW Yun Fat] [HUI On Wah]
```